

DSA 3050A – Business Intelligence & Visualization

Mid-Semester Practical Examination – Business Insights Report

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1. Dataset Overview

The analysis is based on the **v1-legacy-carapi-datafeed** dataset covering model years 2018-2020. After Power Query transformations the final dataset comprises **8,939 vehicle trim records** across **63 columns**, filtered to Model Year ≥ 2018 and Price MSRP > 0 .

Metric	Value
Total Records	8,939
Model Years	2018, 2019, 2020
Brands	~30 (Ford, Chevrolet, Toyota, etc.)
Market Segments	5 (Economy to Ultra-Luxury)
Average Price MSRP	\$50,150
Average Dealer Margin	\$9,250

2. Brand Market Share & Volume

The Clustered Bar Chart reveals vehicle trim distribution across manufacturers:

- **Ford** leads with the highest trim count (~900+), reflecting its broad lineup spanning Entry-Level to Mid-Range.
- **Chevrolet, Toyota, Nissan** rank closely behind — mainstream brands with volume-oriented catalogues.
- **Aston Martin, Alfa Romeo** appear at the low end — consistent with ultra-luxury, low-volume positioning.

Business Implication: Mainstream brands dominate trim variety. Dealers stocking Ford and Chevrolet have higher inventory breadth to serve diverse buyer needs.

3. Pricing Dynamics Across Market Segments

The Clustered Column Chart illustrates the pricing tier structure from Economy to Ultra-Luxury:

Segment	Avg. Price MSRP	Relative Index
Economy	~\$20,000	Baseline
Entry-Level	~\$28,000	1.4x
Mid-Range	~\$45,000	2.3x
Luxury	~\$75,000	3.8x
Ultra-Luxury	~\$200,000+	10x+

Business Implication: The exponential price gap between Ultra-Luxury (~\$200K) and Economy (~\$20K) reflects distinct consumer profiles. The Mid-Range segment at ~\$45K represents an optimal balance of volume and margin.

4. Market Segment Distribution

The Donut Chart and Treemap reveal segment composition by trim count:

- **Mid-Range** is the largest segment (~29%), confirming mass-market appeal in the \$40K-\$55K band.
- **Economy** follows as second-largest (~25%), driven by high-volume entry models from mainstream brands.
- **Ultra-Luxury** and **Entry-Level** are smallest by trim count (~5% each), though Ultra-Luxury commands disproportionate revenue per unit.

Business Implication: Over 50% of the market by trim count sits in Mid-Range and Economy. Businesses should prioritize inventory and marketing spend in these two segments.

5. Model Year Trend Analysis

The Line Chart and Area Chart reveal temporal trends:

- **Trim count declined** from ~3,200 in 2018 to below 2,800 in 2020, reflecting lineup consolidation.
- **Average MSRP trended upward** from ~\$49,000 in 2018 to ~\$51,000 in 2020 (~4% increase), consistent with annual price inflation.

Business Implication: Fewer trim options at higher price points. Dealers should focus on value-added features and financing to maintain volume.

6. Dealer Margin & Profitability

KPI Cards and the Matrix visual provide profitability context:

- **Average Dealer Margin is \$9,250**, approximately 18% of average MSRP.
- Ultra-Luxury brands (Bentley, BMW Ultra-Luxury) carry margins 5-10x higher in absolute dollar terms.
- Mid-Range brands compensate lower per-unit margin through high volume.

Business Implication: Maintain a balanced portfolio — Ultra-Luxury for high-margin transactions, mainstream brands for steady volume revenue.

7. Strategic Recommendations

#	Recommendation	Rationale
1	Prioritise Ford & Chevrolet inventory	Highest trim count and Mid-Range penetration
2	Premium pricing for 2019-2020 model years	Avg MSRP trending up ~4% YoY
3	Focus on Mid-Range and Economy segments	Combined >50% of market by trim count
4	Maintain Ultra-Luxury offerings	Margin 5-10x higher per unit

5	Monitor trim consolidation trends	Trim count declining 2018-2020
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